

STT861 HOMEWORK ASSIGNMENT 01TOTAL: 90 points.Due: 6th Sept, Saturday.

- You can handwrite the solutions. You can also type or write on iPad/laptop etc. Save your homework as a single pdf file and name/save your file as FirstName Lastname_Homework#.
- In D2L, under Assessments, Assignments find the HW# Folder and upload your HW as a single pdf file.
- Points associated is indicated next to each question. Please show detailed work, explain your reasoning clearly and state the results from class or textbook that you are using.
- Please write clearly and legibly using a dark colored pen/pencil.
- This assignment is to be completed individually. Kindly submit your own work. While taking help or borrowing from outside sources (websites, generative AI, etc) is not encouraged, if you do, please be sure to acknowledge and cite the source. Failure to do so will result in a grade of 0 for the entire assignment..

Questions:

1. (10 points) Give a sample space for each of the following situations:
 - a) The number of calls that arrive at a customer service hotline in an hour is counted.
 - b) A customer service representative starts work at 9:00 am and works until 5:00 pm. The elapsed time is measured until the representative receives the first call of his shift.
2. (10 points) For any three events A, B and C, prove that
$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C).$$
3. (20 points) Exercise 1.2 (Part (a) is worth 8 points, the rest 4 points)
4. (30 points) Exercise 1.3 (10 points for each part)
5. (10 points) Exercise 1.10 (5 points each)
6. (10 points) Exercise 1.11 (2 points, 4 points, 4 points)